Kiew Yao Hong

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Summary & Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
Backend: Spring Boot, NestJS, Node.js, Express, FastAPI
Frontend & Tools: React, Astro, Tailwind, Docker, Pivotal Cloud Foundry (PCF), AWS, Git

Experience

DBS Bank *Jul 2024 – Present*
Software Engineer

- Migrated Cards and Payments APIs services spanning Acquiring and Issuing, integrating core banking systems with third-party vendors (VisionPLUS, PowerCARD) to increase in-house ownership and reduce operational costs.
- Designed replay system leveraging ETL pipelines to extract and transform API network traffic for persistent storage, enabling replay-based debugging and reducing turnaround time for diagnosing integration issues.
- Integrated card processing systems with payment schemes (Visa, Mastercard), providing transaction processing capabilities to downstream banking services.
- Led end-to-end change and risk management for the team, overseeing 10+ monthly production changes, ensuring compliance with change governance, audit requirements, risk assessments, approval workflows and post-implementation reviews across regional stakeholders.
- Deployment migration from PCF to OpenShift (OCP), adapting deployment scripts and application configurations to align with the target cloud-native platform.
- Implemented transaction security component for fraud detection API triggered on transaction events to identify and flag unauthorized merchant activity

NCS Group *Nov 2022 – Jun 2024*
Software Engineer

- Modernized legacy systems to cloud-native architecture on AWS, improving reliability, maintainability, and regulatory compliance under Whole of Government (WoG) initiative.
- CI/CD pipeline development and deployment automation, streamlining application releases across development and production environments
- Delivered full-stack web application using React and Spring Boot, refactoring API endpoints, optimizing database queries, and leveraging AWS CloudWatch to reduce incident response time by 50%.
- Led the redesign of React components into reusable, modular patterns, consolidating over 20 static pages into a configurable dynamic module improving Google Lighthouse performance scores from 80 to 90+ while reducing code duplication and improving developer maintainability.
- Proposed and implemented system enhancements aligned with evolving client needs, improving solution adaptability and contributing to long-term revenue growth.

Projects and Miscellaneous

Expense Application | *React, Java, Docker, PostgreSQL*

- Started as a teaching project for an online developer community and evolved into a fully functional finance platform through iterative development.
- Implemented transaction workflows with consistent frontend–backend state synchronization, experimenting with asynchronous event processing to explore system decoupling.

Pathfinding Visualiser | *Python, Pygame*

- Built interactive grid-based graph visualizer supporting dynamic wall placement and start/end node configuration.
- Implemented Dijkstra visualization to animate node exploration and shortest distance

Education

University of Wollongong Singapore Institute of Management
Computer Science, Cybersecurity *Sep 2020 – Mar 2022*